Simon van Lierde

PhD researcher and research software engineer · Open research infrastructure for sustainability science

simonvanlierde@gmail.com · The Hague, Netherlands

github.com/simonvanlierde · linkedin.com/in/simon-van-lierde · simonvanlierde.github.io

PROFILE

I'm a full-stack research software engineer with three years of professional experience in software, data, and solution architecture. I'm now halfway through a PhD in Industrial Ecology at CML, Leiden University. I build open-source tools and data infrastructure that turn environmental research into software people can rely on and find easy to use. As lead developer of R9lab, I own the platform end to end. I also build dashboards and interactive visualizations that make research readable for people outside the field. I work in the open throughout and follow the FAIR principles (findable, accessible, interoperable, reusable).

EXPERIENCE

PhD Researcher in Industrial Ecology @ CML, Leiden University — 2024 – present

- Lead developer of R9lab, responsible for the whole software lifecycle: a FastAPI/PostgreSQL backend with OAuth and Alembic migrations, React/TypeScript and Expo front-ends, CI/CD, Cloudflare networking, restic backups, and security hardening guided by OWASP ASVS.
- Ran two data-collection workshops (135 participants) and deployed the platform in a lab. The pilot recorded 78 products, 1,331 components, and 3,610 images.
- Designed and deployed a central observability stack that CML's research software shares (OpenTelemetry, Grafana, infrastructure managed with OpenTofu).
- Built a Raspberry Pi camera plugin that captures images automatically. It has an OpenAPI spec and is released on PyPI.

Data Manager @ Nationale Milieudatabase (Dutch Environmental Database) — 2023 – 2024

- Managed the environmental databases behind sustainability calculations for the Dutch construction sector. Improved their data structure, consistency, and validation, and advised on software implementation.
- Brought the national environmental databases in line with open data standards (EU LCDN, PEF), working with policymakers, developers, and industry.

Solution Architect & Product Manager @ Metabolic — 2022 – 2023

- Shaped the technical design of tools for LCA (life-cycle assessment) and sustainability reporting, and was their product owner. Guided how the division's projects applied LCA, EEIOA (environmentally extended input-output analysis), and MFA (material flow analysis).
- Product owner and domain expert for automated sustainability-reporting tools (FastAPI, Vue). They replaced static Excel models with REST APIs and web applications.

Consultant & Data Analyst, Sustainable Industries @ Metabolic — 2021 – 2022

- Built environmental-impact models (LCA, MFA) in Python for sustainability consulting projects, with clients in industry, NGOs, and government.

EDUCATION

MSc Industrial Ecology @ Leiden University & TU Delft — 2018 – 2023

Joint degree, specializing in environmental modeling, LCA, and GIS. My thesis used Python geoscripting to model the environmental impacts of space cooling in The Hague.

BSc Physics & Astronomy @ Radboud University — 2014 – 2018

Focus on mathematics, theoretical physics, and programming. Exchange semester at the National University of Singapore.

PUBLICATIONS

Rankin, K. H., Donati, F., Tu, Q., Ward-Bond, J., Reitz, N., Hiser, B., van Lierde, S., Suh, S., Saxe, S., Posen, I. D., & Hawes, J. K. **A call to ensure reproducibility of machine learning applications in industrial ecology**. *Research Square* [preprint]. 2026
10.21203/rs.3.rs-9270723/v1

van Lierde, S.. **R9lab: data collection platform (software, v0.5.0)**. *Zenodo*. 2026
10.5281/zenodo.16637742

PROJECTS

R9lab `github.com/CMLPlatform/r9lab`
Open-source platform for research on taking products apart, the circular economy, and computer vision.

Computer-vision taxonomy visual `simonvanlierde.github.io/cv-ie-taxonomy`
Interactive React/SVG map of a computer-vision taxonomy, drawn as an exploded teardown. A prototype visual to go with an upcoming paper.

MSc-thesis cooling dashboard `simonvanlierde.github.io/msc-thesis-ie`
Interactive dashboard for a model of cooling demand in The Hague and its life-cycle impacts, with Nivo (D3-based) charts and MapLibre maps.

Public web apps `simonvanlierde.github.io`
I maintain a CRediT authorship-statement generator and other open-source web apps. All are tested in CI and deploy automatically to GitHub Pages and Cloudflare.

Open-source contributions
flodym (PIK's material-flow framework): fixed lifetime extension overwriting the model's own parameters, and made it about 9× faster (0.83 s to 0.092 s, 201 years × 12 regions × 8 goods).
lca_algebraic: fixed truncated-normal and lognormal sampling and method lookup, and halved parameter expansion time (0.76 s to 0.40 s at 20k processes).
Activity Browser: kept parameter amounts editable and explained invalid names; loaded scenario files with empty keys and kept a Monte Carlo seed of 0.
Also merged in fastapi-mail (Redis connection handling) and sentier.dev's data tools.

TALKS & TRAINING

- Conference talks: NWO NAC 2026, Brightcon 2025, and ISIE 2025.
- BMVA Computer Vision Summer School (2026)
- HPC Spring School: distributed and GPU computing, Utrecht University (2026)
- Autumn School on inventory modelling at scale, Départ de Sentier (2025)
- PhD Summer School on AI, machine learning and neural networks, Utrecht University (2024)

SKILLS

PROGRAMMING	Python, TypeScript/JavaScript, SQL
FRONT-END	React, Expo/React Native, Nivo (D3), MapLibre, accessibility (WCAG AA, axe-core)
BACK-END	FastAPI, REST/OpenAPI, OAuth, PostgreSQL, Alembic
DEVOPS	Docker, GitHub Actions, OpenTofu/Terraform (IaC), Cloudflare, Grafana, uv, pnpm, devcontainers
PRACTICES	Git/GitHub, pytest, Playwright, PyPI packaging, documentation, reproducible workflows, FAIR and open science, OWASP ASVS
DATA SCIENCE & ML	PyTorch, Pandas
DOMAIN MODELING	LCA, MFA, EEIOA, GIS, circular economy
LANGUAGES	Dutch (native), English (fluent), French (B1), Chinese (A2)

INTERESTS

Climbing · Hiking · Cooking